(X_1 ∧ X_2)∨ X_3

epoch 1

X_1	X_2	X_3	T	$1.0 \times W_0$	$X_1 \times W_1$	$X_2 \times W_2$	$X_3 \times W_3$	Σ	Actual	Alpha × Error	W_0	W_1	W_2	W_3
									Alpha = 0.5		Weight value			
											0.1	0.1	0.1	0.1
1	1	0	1	0.1	0.1	0.1	0	0.3	0	(0.5)(1-0) = 0.5	0.6	0.6	0.6	0.1
0	1	0	0	0.6	0	0.6	0	1.2	1	(0.5)(0-1) = -0.5	0.1	0.6	0.1	0.1
0	0	1	1	0.1	0	0	0.1	0.2	0	(0.5)(1-0) = 0.5	0.6	0.6	0.1	0.6
1	1	1	1	0.6	0.6	0.1	0.6	1.9	1	(0.5)(1-1) = 0	0.6	0.6	0.1	0.6

(X_1 ∧ X_2)∨ X_3

X_1	X_2	X_3	T	$1.0 \times W_0$	$X_1 \times W_1$	$X_2 \times W_2$	$X_3 \times W_3$	Σ	Actual	Alpha × Error	W_0	W_1	W_2	W_3
									Alpha = 0.5		Weight value			
											0.1	0.1	0.1	0.1
1	1	0	1	0.6	0.6	0	0	1.3	1	(0.5)(1-1) = 0	0.6	0.6	0.1	0.6
0	1	0	0	0	0.6	0.1	0	0.7	1	(0.5)(0-1) = -1	0.1	0.6	-0.4	0.6
0	0	1	1	0.1	0	0	0.6	0.7	1	0	0.1	0.6	-0.4	0.6
1	1	1	1	0.1	0.6	0.4	0.6	0.9	1	0	0.1	0.6	-0.4	0.6

(X_1 ∧ X_2)∨ X_3

X_1	X_2	X_3	T	$1.0 \times W_0$	$X_1 \times W_1$	$X_2 \times W_2$	$X_3 \times W_3$	Σ	Actual	Alpha × Error	W_0	W_1	W_2	W_3
									Alpha = 0.5		Weight value			
											0.1	0.1	0.1	0.1
1	1	0	1	0.1	0.6	-0.4	0	0.3	0	(0.5)(1-0) = 0.5	0.6	1.1	0.1	0.6
0	1	0	0	0.6	0	0.1	0	0.7	1	-0.5	0.1	1.1	-0.4	0.6
0	0	1	1	0.1	0	0	0.6	0.7	1	0	0.1	1.1	-0.4	0.6
1	1	1	1	0.1	1.1	-0.4	0.6	1.4	1	0	0.1	1.1	-0.4	0.6

(X_1 ∧ X_2)∨ X_3

X_1	X_2	X_3	T	$1.0 \times W_0$	$X_1 \times W_1$	$X_2 \times W_2$	$X_3 \times W_3$	Σ	Actual	Alpha × Error	W_0	W_1	W_2	W_3
									Alpha = 0.5		Weight value			
											0.1	0.1	0.1	0.1
1	1	0	1	0.1	1.1	-0.4	0	0.8	1	0	0.1	1.1	-0.4	0.6
0	1	0	0	0.1	0	-0.4	0	-0.3	0	0	0.1	1.1	-0.4	0.6
0	0	1	1	0.1	0	0	0.6	0.7	1	0	0.1	1.1	-0.4	0.6
1	1	1	1	0.1	1.1	-0.4	0.6	1.4	1	0	0.1	1.1	-0.4	0.6

Test

(∧)$_V$

Alpha = 0.5

X_1	X_2	X_3	T	$1.0 \times W_0$	$X_1 \times W_1$	$X_2 \times W_2$	$X_3 \times W_3$	Σ	Actual	Alpha × Error	Weight value W_0	W_1	W_2	W_3
											0.1	0.1	0.1	0.1
1	1	0	1	0.1	1.1	-0.4	0	0.8	1	0	0.1	1.1	-0.4	0.6
0	1	0	0	0.1	0	-0.4	0	-0.3	0	0	0.1	1.1	-0.4	0.6
0	0	1	1	0.1	0	0	0.6	0.7	1	0	0.1	1.1	-0.4	0.6
1	1	1	1	0.1	1.1	-0.4	0.6	1.4	1	0	0.1	1.1	-0.4	0.6
0	1	1	1	0.1	0	-0.4	0.6	0.3	1	0	0.1	1.1	-0.4	0.6
1	0	1	1	0.1	1.1	0	0.6	1.8	1	0	0.1	1.1	-0.4	0.6

Accuracy : $\frac{TP + TN}{(N)} = \frac{1 + 1}{2} = 1$

Precision : $\frac{TP}{TP + FP} = \frac{1}{1+0} = 1$

Recall : $\frac{TP}{TP + FN} = \frac{1}{1+0} = 1$

Specificty : $\frac{TN}{TN + FP} = \frac{1}{1+0} = 1$